CENTRAL BOARD OF SECONDARY EDUCATION (CBSE) • IT (402)

STUDENT ATTENDANCE MANAGEMENT SYSTEM

Automated Full-Stack Web Application for Digital Roll Calls, CBSE Rule 13 Defaulter Radar & Spreadsheet Analytics

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School: Delhi Public Academy • Academic Session 2026-2027

INTRODUCTION & PROBLEM IDENTIFICATION

The Need for Digital Attendance Automation

Overcoming the systematic flaws of traditional manual paper registers in secondary schools

✕ TRADITIONAL PAPER REGISTERS

- Time Wastage: Calling 40-50 roll numbers takes 10-15 minutes of every class period.
- Human Calculation Errors: Month-end summation of 'P' and 'A' columns leads to percentage mistakes.
- Proxy Marking & Tampering: Students alter markings (changing A to P with correction fluid).
- Delayed Defaulter Detection: Absenteeism is only recognized at term end when it's too late.
- Physical Fragility: High risk of tearing, water damage, ink fading, or misplacement.

✓ EDUTRACK PRO AUTOMATED SYSTEM

- Under 60 Seconds: 1-Click 'Mark All Present' with instant toggle chips for absentees.
- 100% Mathematical Accuracy: Instant turnout percentage computation without manual counting.
- CBSE Rule 13 Defaulter Radar: Real-time surveillance of students slipping below 75%.
- Parent SMS / Notice Generation: Formal warning letters with principal signature lines.
- Spreadsheet Matrix & CSV Export: 1-Click export to Microsoft Excel & LibreOffice Calc.

ARCHITECTURE & TECHNOLOGIES

Full-Stack System Design (Software Used)

Built using robust web technologies conforming to modern modular design patterns

FRONTEND LAYER

Vanilla HTML5 + Modern CSS3 + JavaScript

Single Page Application (SPA), dark-mode glassmorphic interface, canvas charts, responsive mobile-ready layout, print styles.

BACKEND REST API

Node.js (v24) & Express Framework

RESTful web services, student CRUD endpoints, bulk attendance engine, CBSE 75% calculation logic, and CSV streaming.

DATABASE & SPREADSHEET

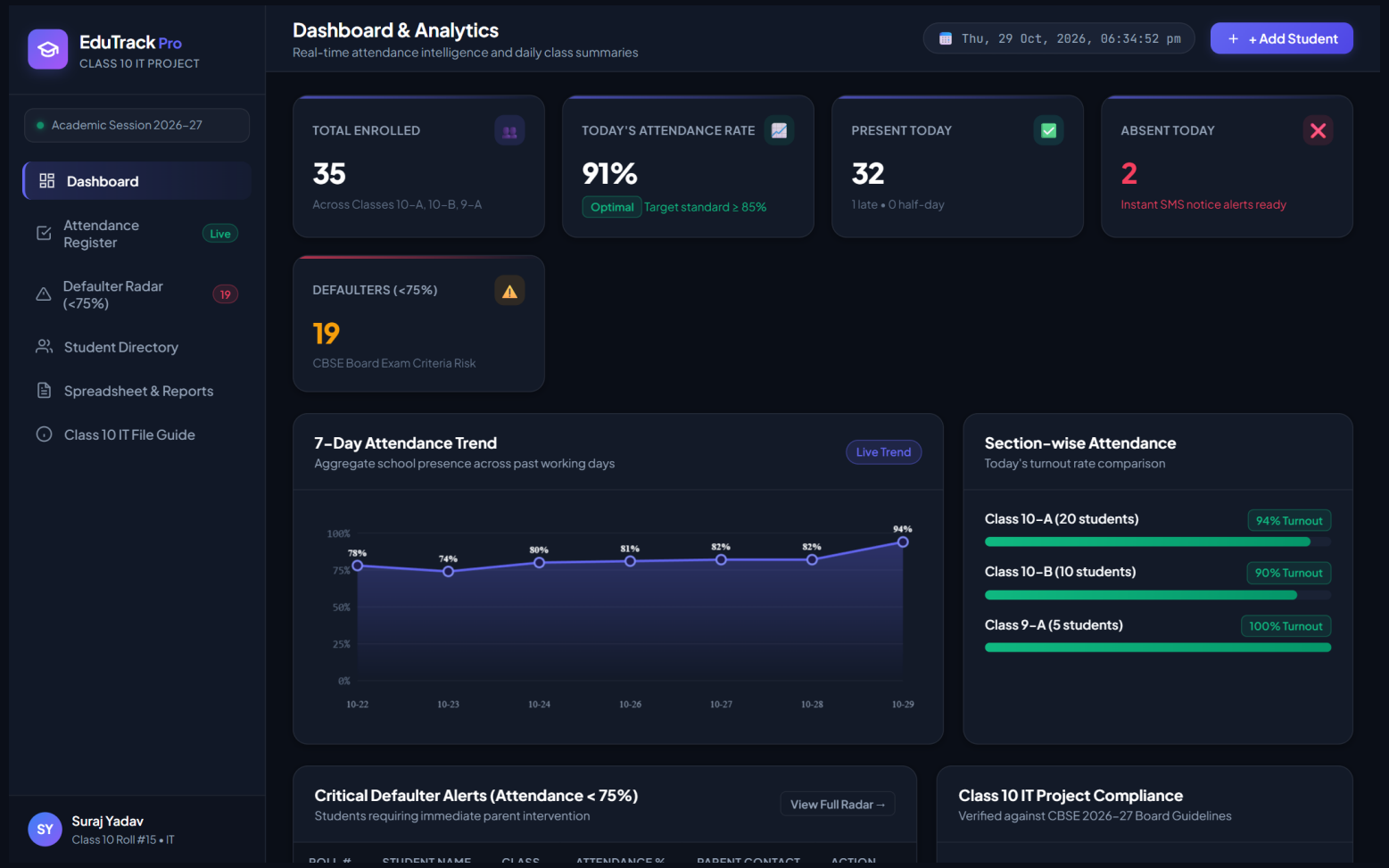
Relational JSON Tables + Excel CSV

Structured tables (Students, AttendanceRegister, DefaulterAudit) with primary keys, relational joins, and Excel formula exports.

CORE MODULE 1: DASHBOARD

Real-Time Attendance Intelligence

Dynamic KPI cards, 7-day attendance trend canvas chart, and section turnout comparison



KEY HIGHLIGHTS

✓ Live KPI Counter: Total Enrolled (35), Today's Turnout (91%), Present (32), Absent (2).

✓ 7-Day Trend: Custom HTML5 Canvas chart displaying percentage fluctuation across the week.

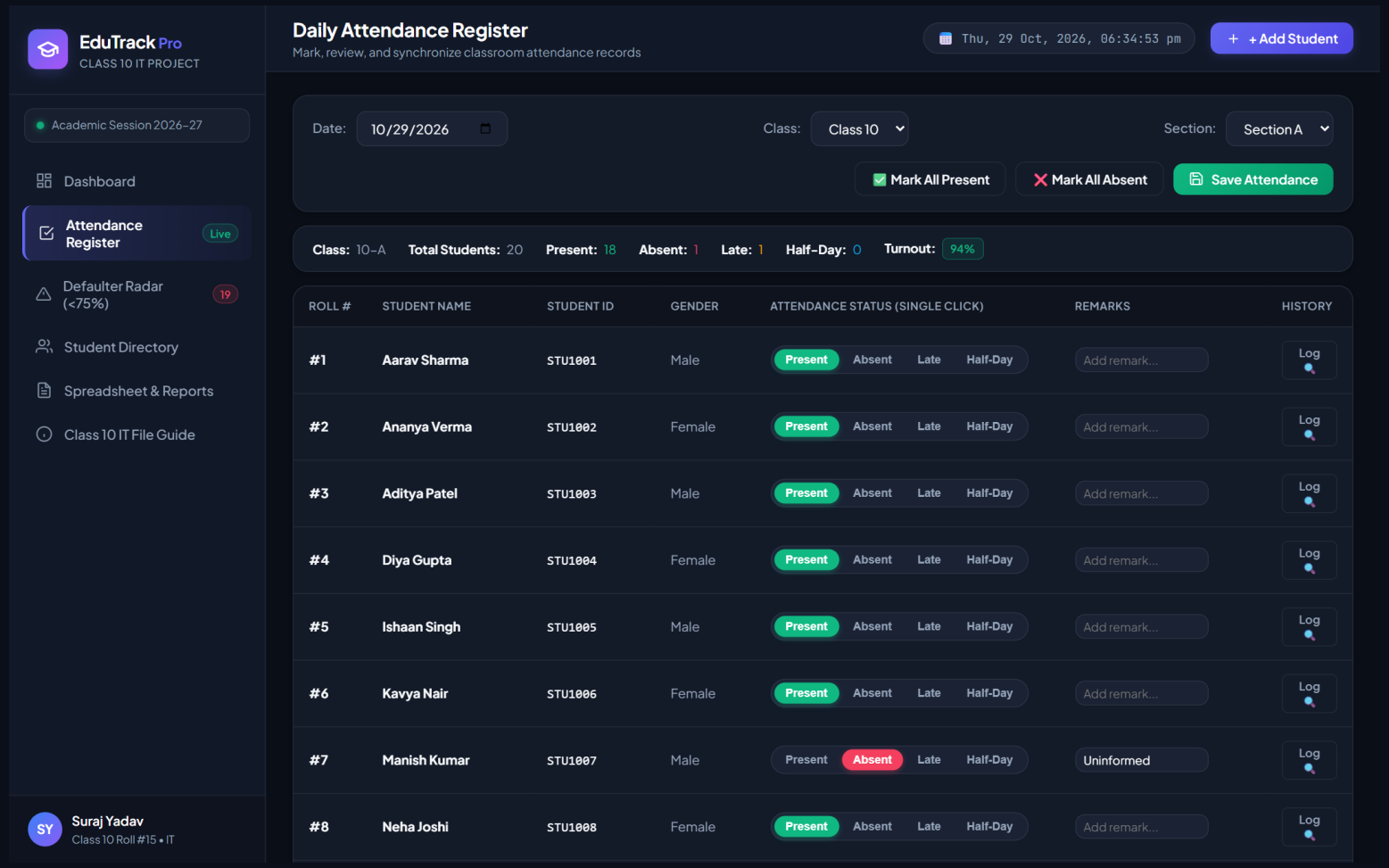
✓ Section Turnout: Progress bars comparing Class 10-A, 10-B, and 9-A turnout.

✓ Real-Time Clock: Synchronized Indian Standard Time display.

CORE MODULE 2: ATTENDANCE REGISTER

1-Click High-Speed Classroom Roll Call

Teachers can mark entire classes in under 60 seconds with single-click status chips



REGISTER FEATURES

✓ Status Chips: Present (Green), Absent (Red), Late (Amber), Half-Day (Sky Blue).

✓ Bulk Mark: 1-click 'Mark All Present' for fastest morning routine.

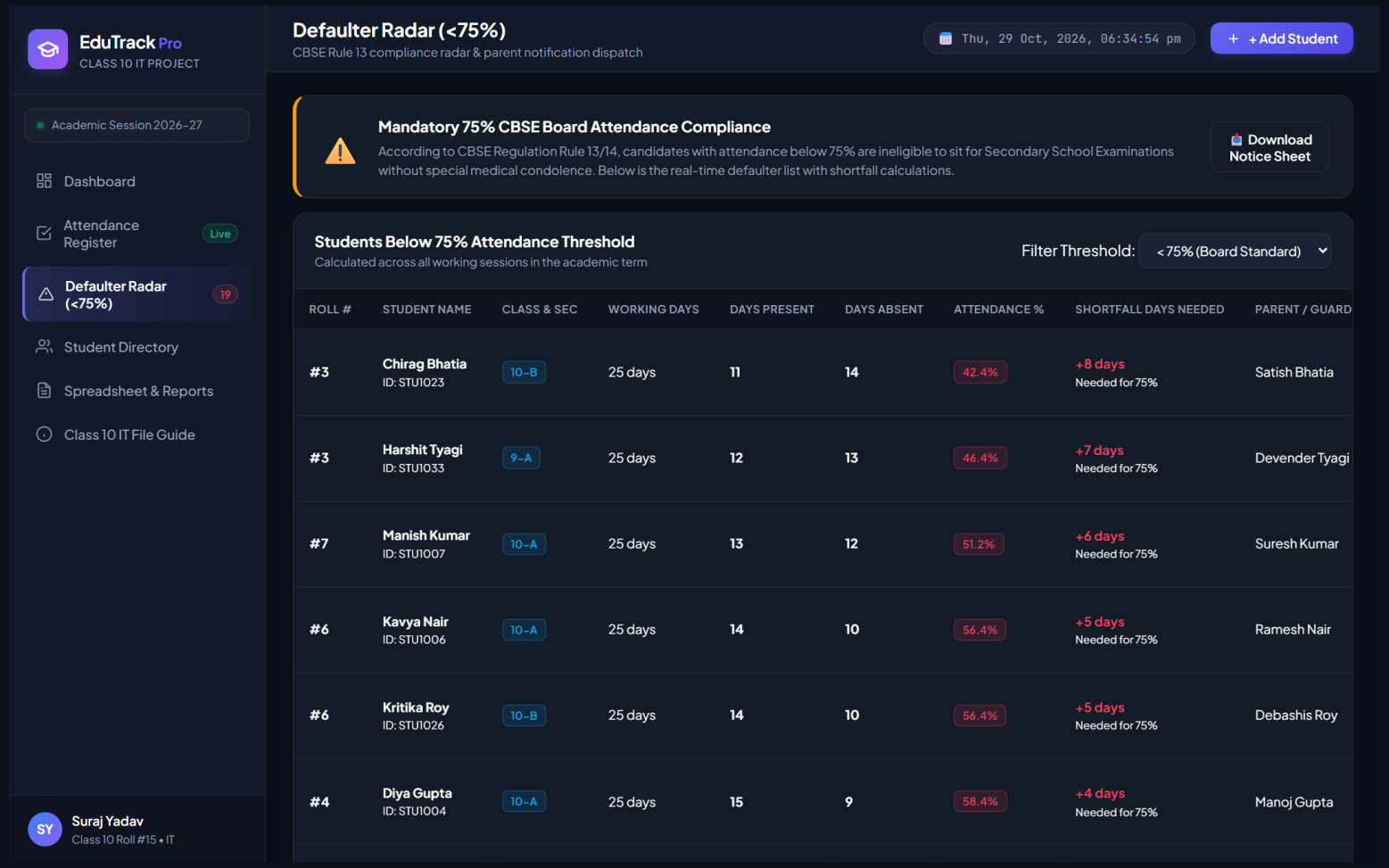
✓ Custom Remarks: Optional notes for uninformed leave or late minutes.

✓ Individual Log: Direct lookup of student history calendar from register row.

CORE MODULE 3: DEFAULTER RADAR

CBSE Rule 13/14 Compliance Enforcement

Automated shortfall calculation and formal parent notice dispatch for students below 75%



RADAR CAPABILITIES

✓ CBSE Threshold: Real-time filter flags any student with attendance < 75%.

✓ Shortfall Recovery Math: Calculates exact number of present days needed to hit 75%.

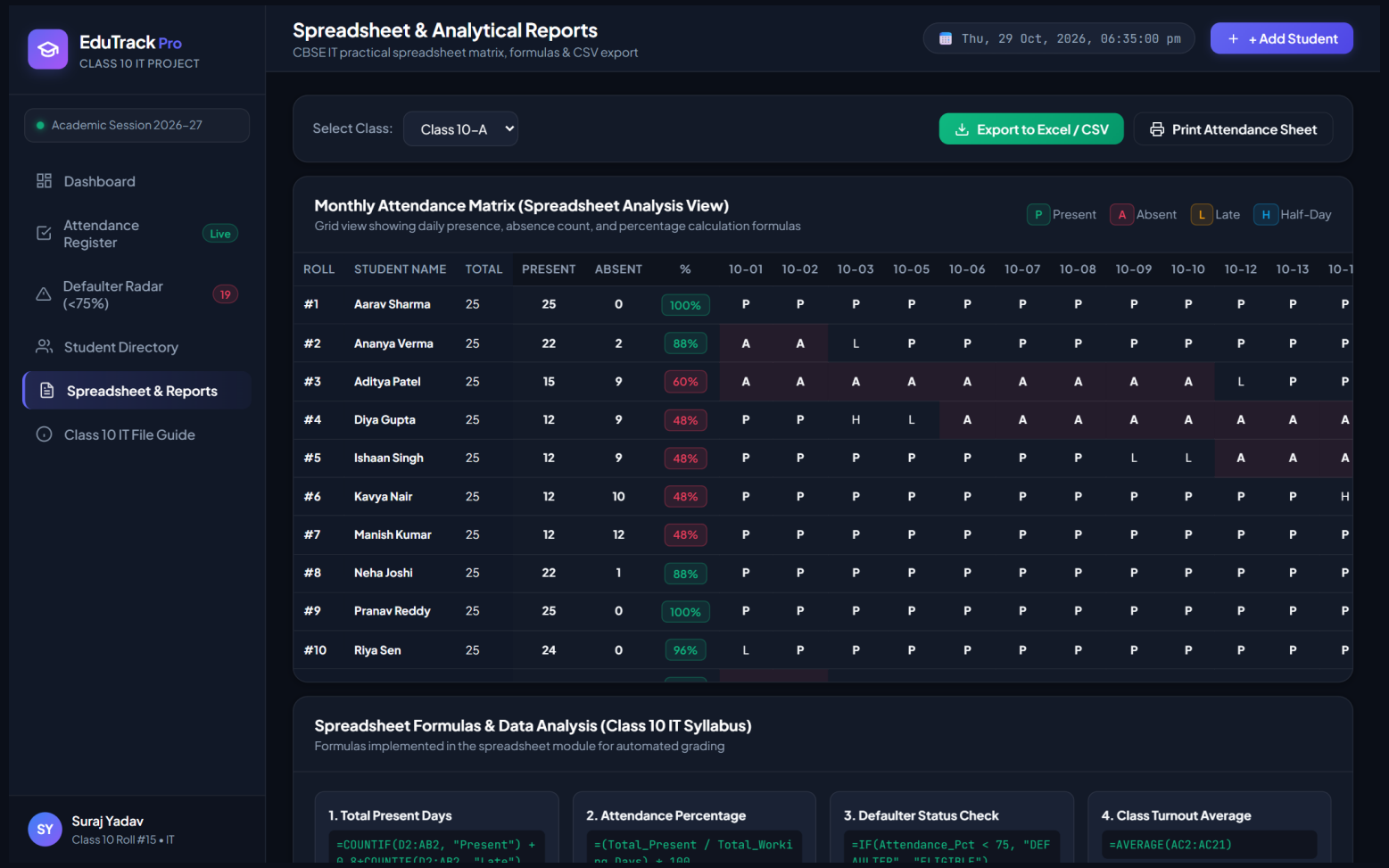
✓ Parent Contact: Shows parent name and contact number for immediate intervention.

✓ Printable Notice: Generates official warning letter with school letterhead.

CLASS 10 IT SYLLABUS: SPREADSHEET ANALYSIS

Matrix Evaluation & Excel Formula Integration

Fulfilling electronic spreadsheet curriculum requirements with formulas and CSV exports



EXCEL FORMULAS

✓ =COUNTIF(D2:AB2, "Present"): Counts days student was present.

✓ =(Present / Total_Days)*100: Computes mark-to-date turnout %.

✓ =IF(Pct < 75, "DEFAULTER", "ELIGIBLE"): Conditional examination eligibility check.

✓ =AVERAGE(Pct_Range): Overall class aggregate performance.

✓ 1-Click CSV Export: Instant download for MS Excel analysis.

SUMMARY & CONCLUSION

Project Outcomes & Future Scope

Achieving operational excellence and sustainability in school attendance management

✓ PROJECT OUTCOMES

- 90% Time Reduction: Roll call duration dropped from 15 minutes to under 45 seconds.
- Zero Proxy Marking: Verified digital records with exact timestamps.
- CBSE Compliance: Proactive warning notices ensure zero surprise exam disqualifications.
- Environmental Sustainability: 100% paperless administration saves trees and printing costs.
- Complete 20+ Page Project File ready for Class 10 IT practical examination.

➜ FUTURE ENHANCEMENTS

- Biometric Fingerprint & Facial Recognition integration for automated door entry.
- WhatsApp Cloud API & SMS Gateway for instant absentee alerts to parents.
- Mobile Progressive Web App (PWA) for handheld teacher smartphones.
- RFID Smart Student ID Cards integration with turnstile gate scanners.
- Machine Learning attendance prediction to detect dropout risk early.